



INTRODUCTION TO

engineering &
THE **environment**

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chapter

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Engineering and the Environment



1.1 INTRODUCTION

Traditionally, the study of engineering has divided along disciplinary lines, most commonly electrical, chemical, civil, and mechanical engineering. From time to time, more specialized topics of popular interest have emerged, giving rise to new fields such as aerospace engineering (an outgrowth of mechanical engineering, popular in the 1960s) and computer engineering (an outgrowth of electrical engineering, popular in the 1980s and 1990s). Environmental engineering—the study of engineering solutions to environmental problems—is also widely seen as a specialized area gaining in popularity.

Historically, the field has been largely identified with civil engineering, where environmental studies have focused primarily on the problems of water pollution and solid waste disposal. Today, however, environmental problems extend far beyond the traditional domains of civil engineering alone. Indeed, the premise of this book is that solutions to modern environmental problems are part of *every* engineering discipline. No engineer can think that environmental problems will be solved by “someone else.” As we shall see, nearly everything we do as engineers can have an impact on the environment—impacts that can be part of the solution or part of the problem.

This chapter highlights some of the ways in which engineers of all stripes are involved in both creating and solving environmental problems. Although you may not think of yourself as an environmental engineer, you may be surprised to see the many links between engineering and the environment, no matter what your field of interest.

1.2 WHAT IS “THE ENVIRONMENT”?

Most people care about the environment. But what exactly does that mean? When we talk about “environmental problems” that need to be addressed, what do we have in mind?

The “environment” can mean different things to different people. The unabridged Random House dictionary defines *environment* as “the aggregate of surrounding things, conditions or influences, especially as affecting the existence or development of someone or something.” What we have in mind when talking about the environment can make a big difference in the way we approach the subject and in how we formulate or frame environmental issues and problems.

In this book the term *environment* will generally refer to the physical environment that surrounds us. This includes the air we breathe, the water we drink, and the lands, oceans, rivers, and forests that cover the earth. To an increasing extent it also includes the buildings, highways, and modern infrastructure of the urban settings in which a growing proportion of the world’s population resides. The state of this physical environment directly and indirectly affects the viability of all living things on the planet—the people, plants, birds, fish, and other animals that we care about. The welfare of these living things motivates most environmental concerns.

This chapter provides a brief introduction to some of the major environmental themes that are discussed later in the text. These themes emphasize areas where engineers play a major role in solving (as well as creating) environmental problems.

For perspective, we begin with a broad overview of environmental impacts and the role of engineering.

1.3 FRAMING ENVIRONMENTAL ISSUES

We know from the study of geology, biology, and other natural sciences that the earth's environment has been changing since time began. What motivates modern concerns about the environment is the expanding role that human activities play in accelerating environmental change.

Figure 1.1 illustrates schematically how human activity influences the environment. At the most basic level, human demands for food and shelter mean that some living things (plants and animals) will be killed or harvested for food, and some natural resources, such as trees, will be used to build structures and provide energy for shelter, cooking, and warmth. These basic demands of human survival thus give rise to an environmental impact. As human populations increase, more and more land is altered to provide settlements and to support activities such as agriculture, industrial processes, and transportation systems. And as people become more affluent, their demand on the earth's resources grows far beyond the basic needs for survival.

In addition to altering the landscape, diverse human activities give rise to various types of waste emissions or *residuals* discarded to the environment. Modern examples include air pollutants from factories and automobiles, water contaminants from manufacturing processes, solid wastes from household and municipal activities, and pesticides used in agriculture.

Many of the wastes or pollutants discharged to the environment are subsequently transported and chemically or biologically transformed over time and dis-

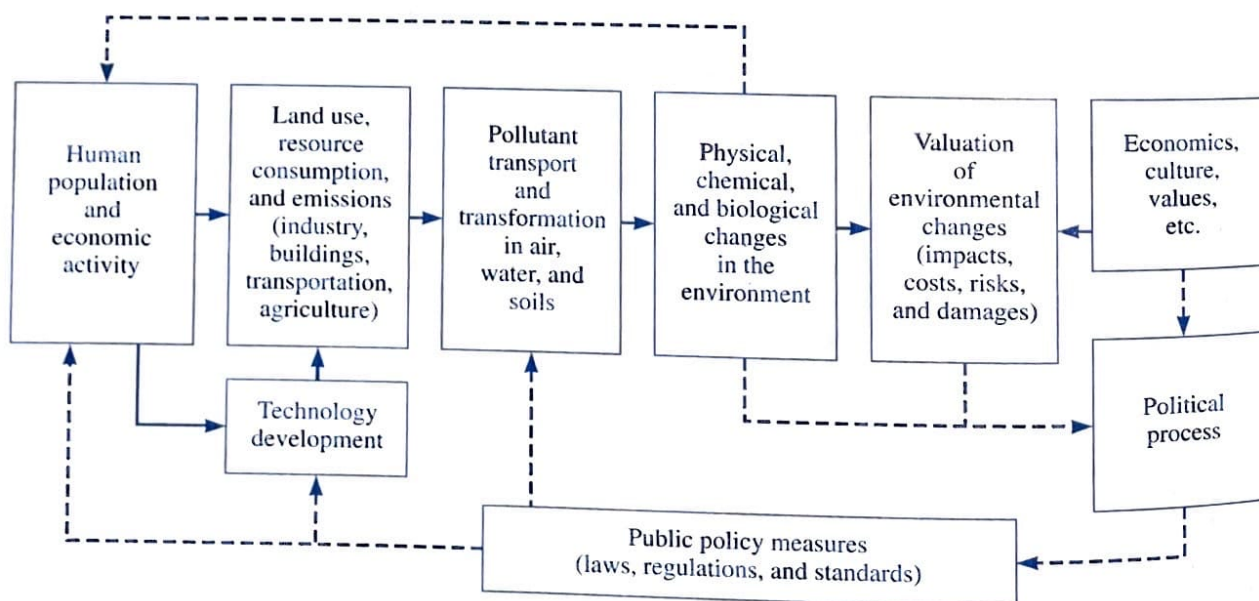


Figure 1.1 A framework for environmental impact assessments. Solid lines show the path of primary or initial impacts; dashed lines show the major feedbacks and responses to these impacts.

tance. For example, gaseous sulfur dioxide emitted from the tall chimneys of coal-burning power plants is partially converted to fine sulfate particles by chemical processes in the atmosphere. This mix of sulfurous gases and particles leads to *acid rain*, which can damage lakes and forests hundreds of miles away. Similarly, many types of organic wastes from factories and households are discharged to rivers and streams, where they are transformed by biological reactions that deplete the oxygen dissolved in water. A deficiency of oxygen directly affects the viability of all aquatic species. Other examples of physical, chemical, and biological changes in the environment due to human activities are shown in Table 1.1.

Historically, the most pervasive environmental changes caused by people have been related to land use—particularly the clearing of forests for agriculture and urbanization. Now, in the modern industrial era, the added impacts of emissions from modern industrial technology are rapidly accelerating the process of human-induced environmental change.

1.3.1 Good Change or Bad?

Are all changes in the environment a reason for concern? Isn't the environment always changing, and aren't many of these changes beneficial?

The answers to such questions are not always simple. In general, our greatest concern is over environmental changes that may harm us or affect our welfare. This is illustrated by the feedback loop in Figure 1.1 between environmental change and human activities. This interaction is complex because some environmental changes that are beneficial in the short run may have adverse consequences later on. In preindustrial

Table 1.1 Some examples of environmental change from human activities.

Human Activity	Physical Changes	Chemical Changes	Biological Changes
Land and water use for housing, agriculture, industry, transportation, and recreation	Deforestation and other alterations of landscapes (e.g., changes in terrain slope, vegetation coverage, pavement); alteration of waterways (e.g., flooding, dams, changes in river channels, drainage of wetlands)	Changes to chemical constituents of soils and sediments (e.g., increased acidity and turbidity of waters, removal of nutrients from soils)	Changes in the viability of plants, fish, animals, and microorganisms due to altered habitat and chemical constituents or concentrations, possibly leading to species succession, extinction, migration, or disease
Emissions or discharge of chemical substances to air, land, and water	Changes to the built environment (structures such as buildings, bridges, monuments, etc.) from deposition and chemical attack caused by emissions such as soot deposits, acid gases, and liquid chemicals	Increases in the concentration of emitted substances in the air, water, and soil; other chemical changes resulting from secondary reactions (e.g., ozone buildup in urban areas)	Injury or illness to people, plants, and animals from exposure to and/or accumulation of chemicals and their derivatives



Land use impacts of human activities. This countryside in southern California originally was transformed for agriculture. Now tree groves have been uprooted to make way for construction of roads and expansion of human settlements.

societies, for instance, the clear-cutting of forests to support agriculture was essential for providing food for a growing population. Over time, however, the continued depletion of forests often led to soil erosion, loss of soil nutrients, and a subsequent inability to sustain agricultural production. As a result, many communities disappeared or were forced to migrate elsewhere.

In today's industrial society, many air pollutants and water contaminants entering the environment as byproducts of modern technology are widely recognized sources of human illness and ecological damage. More subtle or indirect impacts, such as the effects of long-term global warming from anthropogenic emissions of carbon dioxide and other gases, are less clearly defined at the moment but raise new concerns about the longer-term effects of energy use and industrial activities.

Of course, not all people agree on whether a particular environmental change is cause for concern. For example, the clearing of wooded lands to provide space for a new farm, factory, shopping mall, or housing development is commonly seen as a beneficial change that allows people to lead better, more productive lives. Others, however, may view these same changes as destructive of natural habitats for vegetation and wildlife that have intrinsic value and are part of natural ecosystems that affect our long-term welfare and survival. So who's right? And if both views have merit, how do we weigh the tradeoffs and make a decision?

1.3.2 Enter Public Policy

In democratic societies, the political process is how decisions are made as to whether actual or potential changes in the environment are of sufficient concern to adopt *policy measures* (e.g., laws and regulations) to prevent, alter, or reverse these changes. Some notion of the *damages* or *risks* that will be avoided by taking action is one essential element of the decision-making process. Providing such information is an area where science and engineering often play a key role. Ultimately, a host of other factors also influence community and national decisions about environmental poli-

cies. As shown in Figure 1.1, such factors include the influences of culture, societal values, and economics.

Because of these factors, priorities and preferences for environmental protection often vary from community to community and from country to country. For instance, a nation struggling to provide its citizens with the basic necessities of life is unlikely to be as worried about wilderness preservation as a wealthy nation. Nonetheless, over time societies tend to address environmental problems in an order roughly equivalent to the relative risks they pose. Often this starts with basic health issues like water quality and sanitation, then broadens (often in tandem with economic development) to include other issues of health and environmental quality.

Environmental policies also are often based on concepts of fairness, or equity, such as the idea that all citizens have a right to breathe clean air. In other cases, the symbolism or basic ethic of environmental protection is most important, as with the protection of endangered species. Because environmental policy often has significant economic implications, it is almost always influenced by private interests as well as the public interest.

At all stages in the policy development process, one important approach is to look for policies whose benefits clearly outweigh the cost of measures taken. An economic analysis of costs and benefits is sometimes employed to evaluate the merits of proposed policy measures. In many cases, however, putting a dollar value on expected environmental and health benefits is especially difficult and controversial. A less controversial approach is to identify measures that are lowest in cost. The adoption of such measures often serves as an initial basis for policy actions.

If environmental protection measures are adopted, their influence might be felt in a number of ways. For example, some measures might alter human behavior by forbidding certain types of activities, such as the dumping of wastes or drilling for oil in pristine areas. Other measures might require the use or development of new technology to abate harmful emissions to the environment. Thus, new cars might have to be equipped with catalytic converters to reduce air pollution and cleaner technologies may be demanded to generate electricity. To an increasing extent, environmental protection measures in the form of standards and regulations have established new constraints on the design and deployment of modern technology. Environmental policies thus shape the development of technology in directions that reflect the goals and preferences of society.

1.4 THE ROLE OF ENGINEERING

Where do engineers fit into all of this? Figure 1.2 suggests how pieces of the environmental puzzle map roughly into the traditional bailiwicks of undergraduate majors or disciplines. Engineers are primarily involved in problems related to technology development and deployment. These include designing, developing, and building the cars, computers, television sets, and other consumer products that people enjoy. Engineers also design and build all the manufacturing processes, industrial technology, and transportation infrastructure needed to extract, transport, and refine raw materials; fabricate products; and distribute the goods and services of

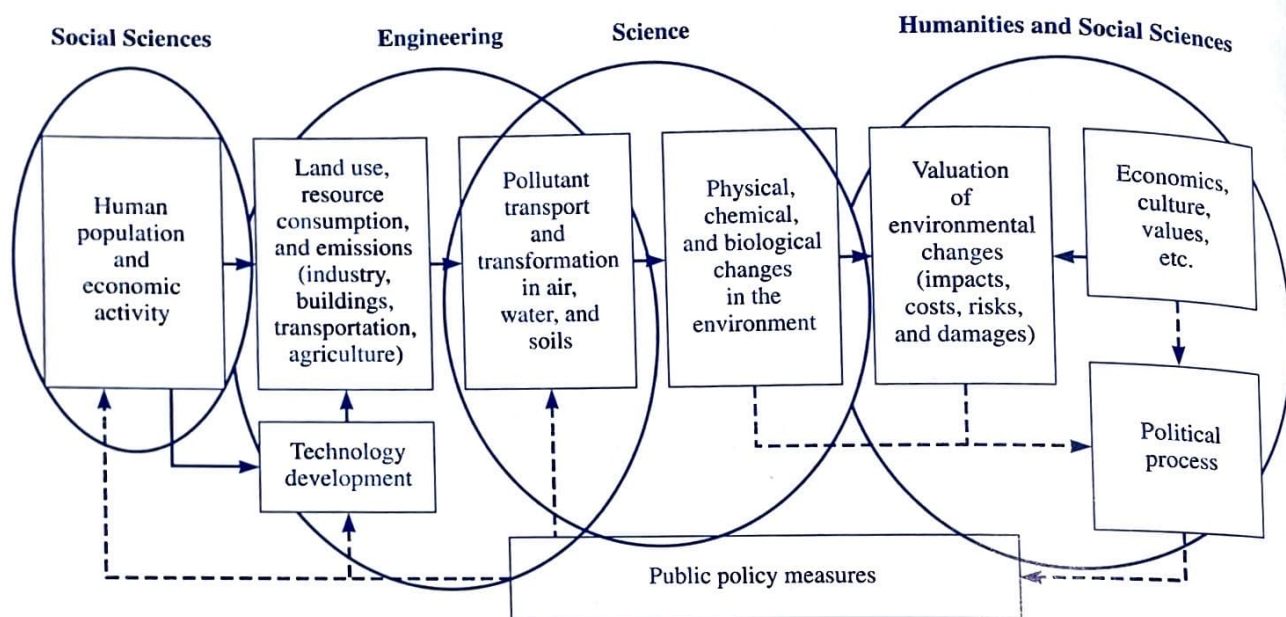


Figure 1.2 Mapping of environmental topics into traditional undergraduate disciplines.

modern societies worldwide. In order to predict the consequences of technology deployment, engineers, along with scientists, also are involved in the study of how pollutants are transported and transformed in the environment. Thus, in the broadest sense, engineers are concerned with—and often responsible for—a wide range of activities that directly or indirectly contribute to environmental change.

From this discussion we see that the sources of anthropogenic environmental change fall into two broad categories: (1) *changes associated with land use* (including depletion of natural resources) and (2) *changes induced by emissions* or residues from products and industrial processes. Engineers have responsibilities for both types of impacts, especially the latter. For example, chemical and mechanical engineers design the petrochemical refineries, paper mills, power plants, and myriad other processes at the heart of industrial societies. The design and operation of these processes directly determine how much air pollution, water pollution, and solid wastes are produced and released to the environment. The same is true for the water treatment plants designed by civil engineers, the microchips developed by electrical and computer engineers, and the steel mills, aluminum smelters, and extraction processes developed by metallurgical engineers.

In addition, all of the consumer products we use (and eventually discard)—such as batteries, toasters, light bulbs, refrigerators, computers, and automobiles—are the result of engineering design decisions that impact the environment. For example, the engineering design of an automobile, including the choice of materials, the type of engine or power plant, and the manner of construction, determines the type and level of air pollutants that are emitted over the life of the car, the amount of fuel required, and the type and quantity of solid wastes that must be disposed of at the end of the car's useful life. Thus, the way engineers design products and processes plays a key role in creating as well as solving environmental problems.

What about other types of environmental concerns, such as species extinction from the clearing of tropical forests, drilling for oil in the Arctic, or the ecosystem impacts of wetlands development? Do engineers also have a professional role in these types of issues?

Yes, they do. These are examples of environmental changes associated with land use. Here engineers also play a key role, but often less directly. The environmental impact of land use decisions is a broader societal issue that involves not only engineers, but a host of other professional and political interests as well. Thus, while the technical design of an automobile, a power plant, or a steel mill is primarily an engineering responsibility with direct consequences for emissions to the environment, decisions about *deploying* that technology—that is, *whether* to use it, or *where* to put it, or *how many* to deploy—typically extend beyond the realm of engineering practice. Often those decisions fall in the domains of city and regional planners, government regulatory agencies, and other groups responsible for review and approval of land use proposals from developers, landowners, corporations, and others. Thus, although engineers frequently play a central role in formulating recommendations with respect to land use—such as where to locate a highway, build a dam, or deploy an oil rig—the process of arriving at a final decision transcends engineering. Throughout this process, however, engineers are intimately involved in defining, collecting, and interpreting the data needed to assess the environmental implications of major land use decisions.¹

1.5 APPROACHES TO “GREEN” ENGINEERING

Think about something you’ve thrown away recently. Maybe it was something small like the alkaline batteries from your CD player, or something big like an old refrigerator or an old used car that finally died. Now imagine that you were the engineer responsible for designing and manufacturing that product. The laws have suddenly changed, and now you must take back all of the discarded product and be responsible for its environmentally safe disposal. If you had known this in advance, would you have designed and built the product in the same way?

This scenario is not at all far-fetched. In Germany, manufacturers like IBM are now required to take back all of their discarded computers, which contain quantities of toxic metals used to manufacture circuit boards and other electronic components. The cost of dealing with this enormous problem has caused IBM to completely rethink the way it designs, manufactures, and ships its computers. The result has been a substantial reduction in environmental impacts throughout the product life cycle, without compromising the product’s performance or adding to its cost. Indeed, many changes have *saved* money.

This is but one example of how the philosophies of green design, pollution prevention, and industrial ecology are changing the way engineers do their jobs. Here

¹ Note that the term *land use* as used here should be thought of broadly to include all of the earth’s surface, including oceans as well as land. Indeed, as we enter the 21st century, our sense of the natural environment may expand to include outer space. Over the past 30 years, for example, an increasing array of high-tech debris from satellite launches and other space program ventures has been left in orbit as “space junk.” Might space pollution be the next environmental frontier?